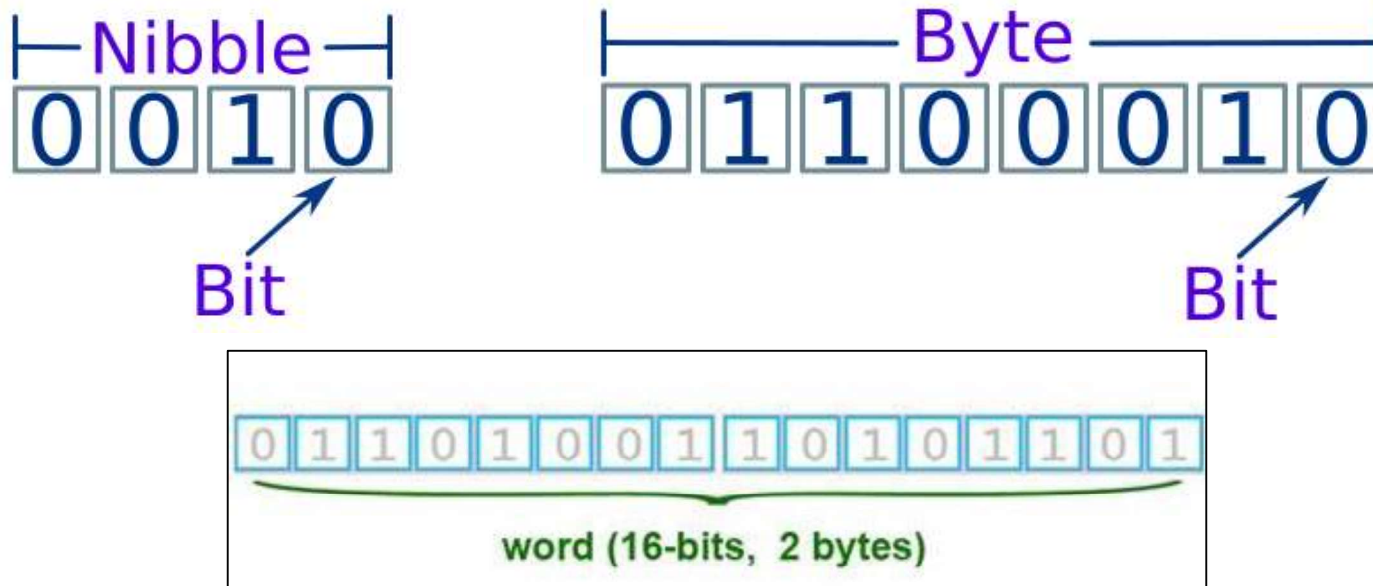


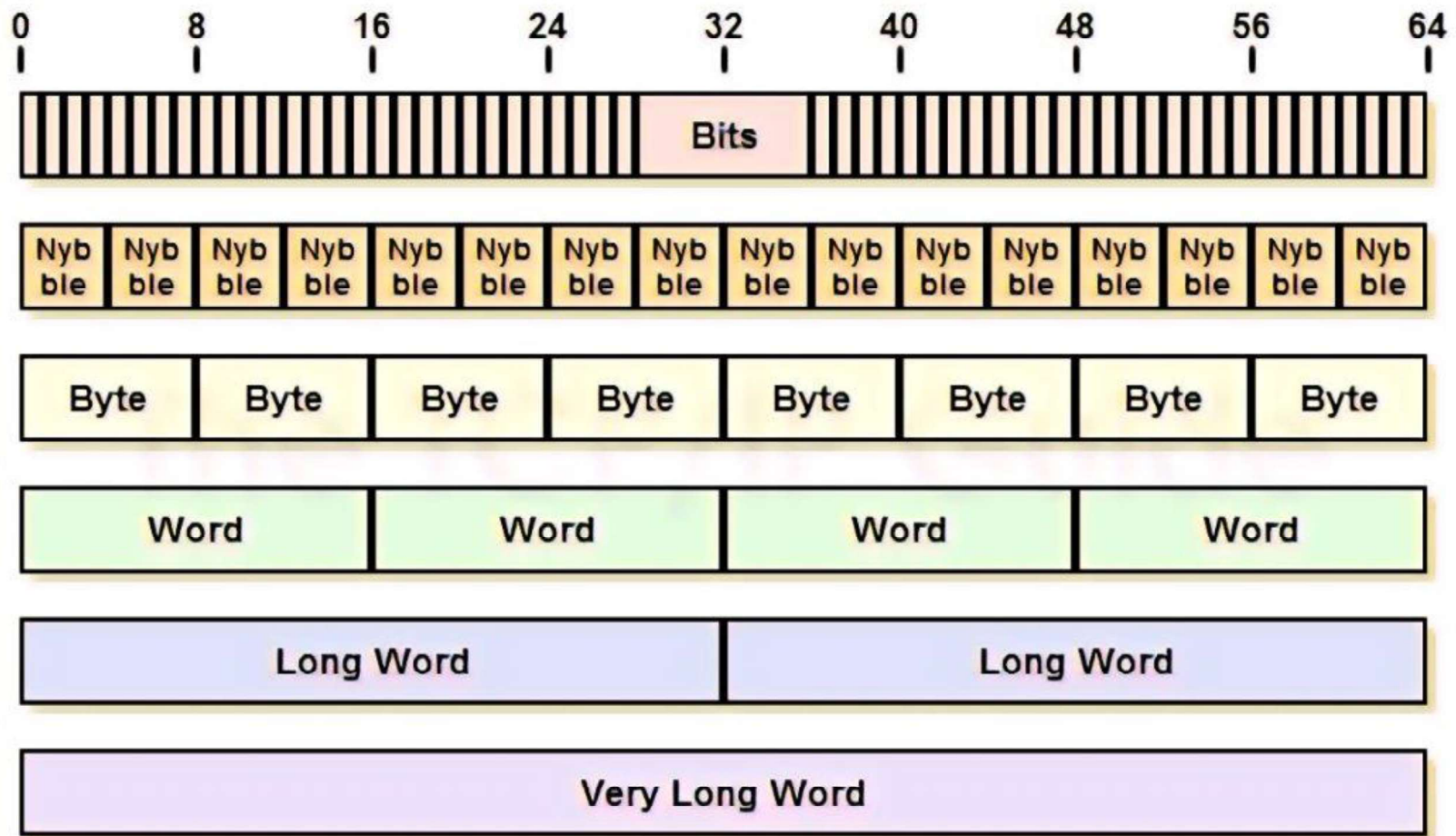
Introduction to Computing (ITC)

CS1101

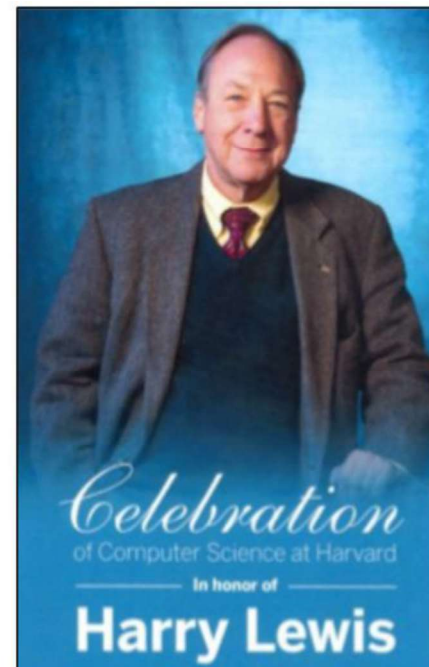
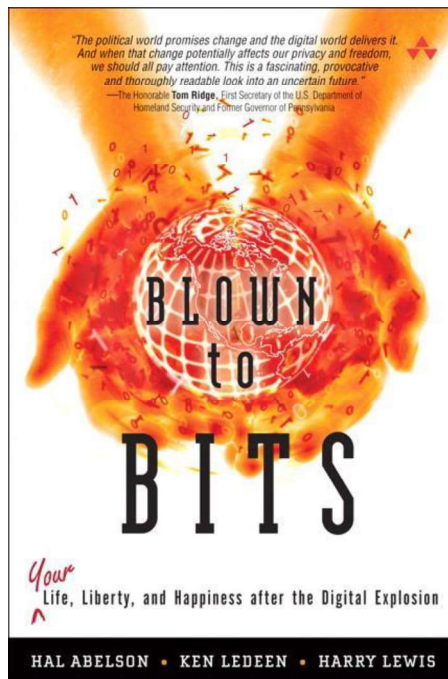
Lecture-7

Bit, Nibble, Byte, Word, long-word

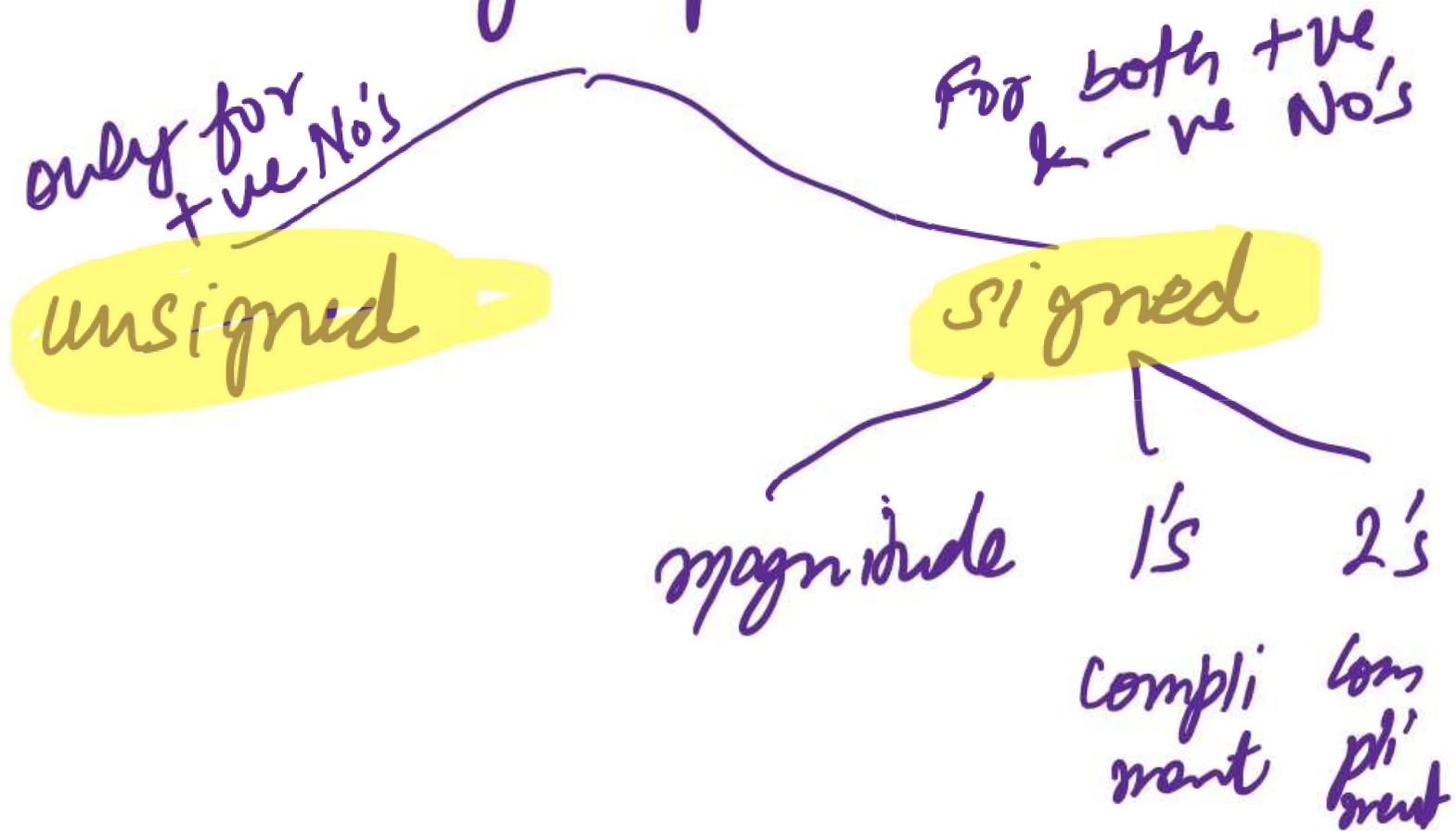




Bits, Bits everywhere



Binary representation



What are the issues with representing Binary data?

- Numeric data
- Positive integer

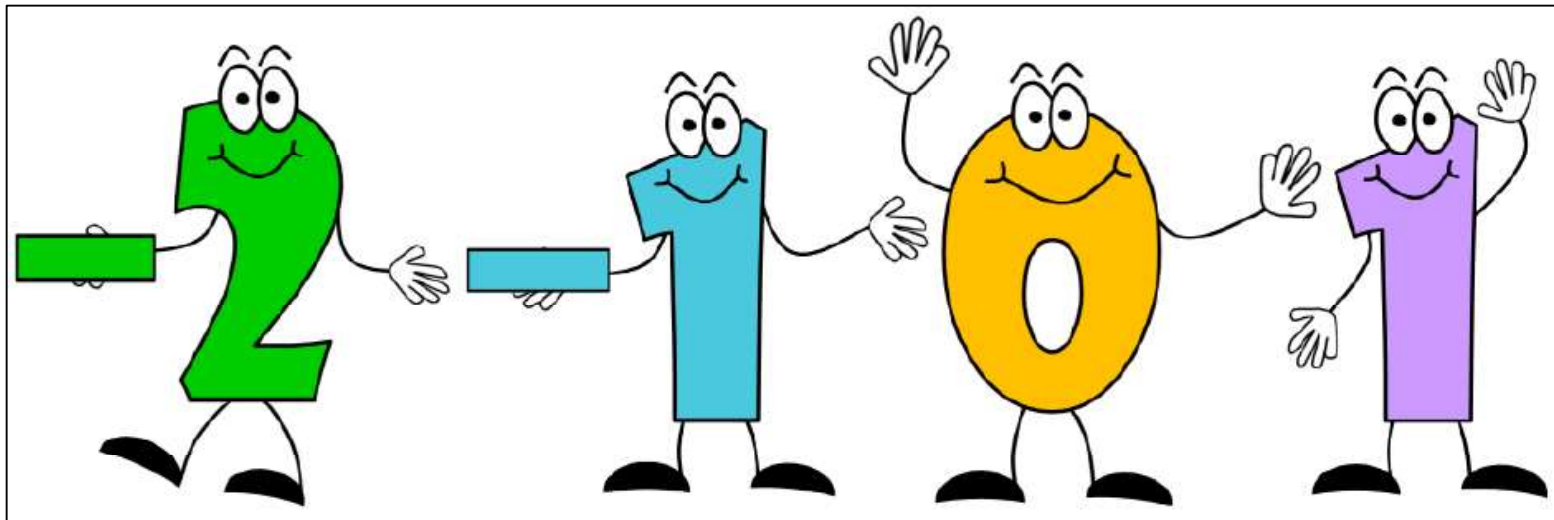


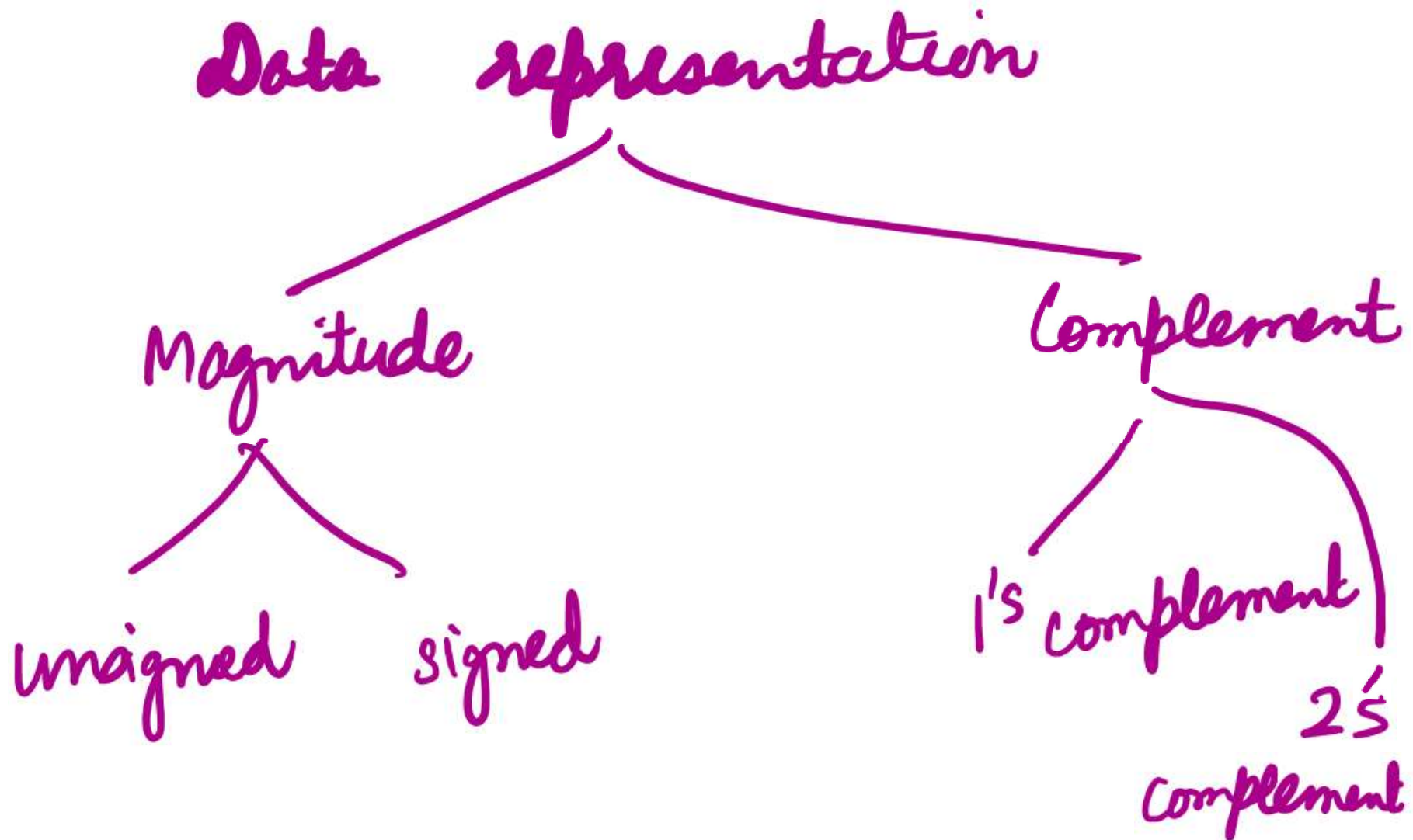
- **What about Negative/Non-integer values?**

-6?, -5, -2.3,.....and so on?

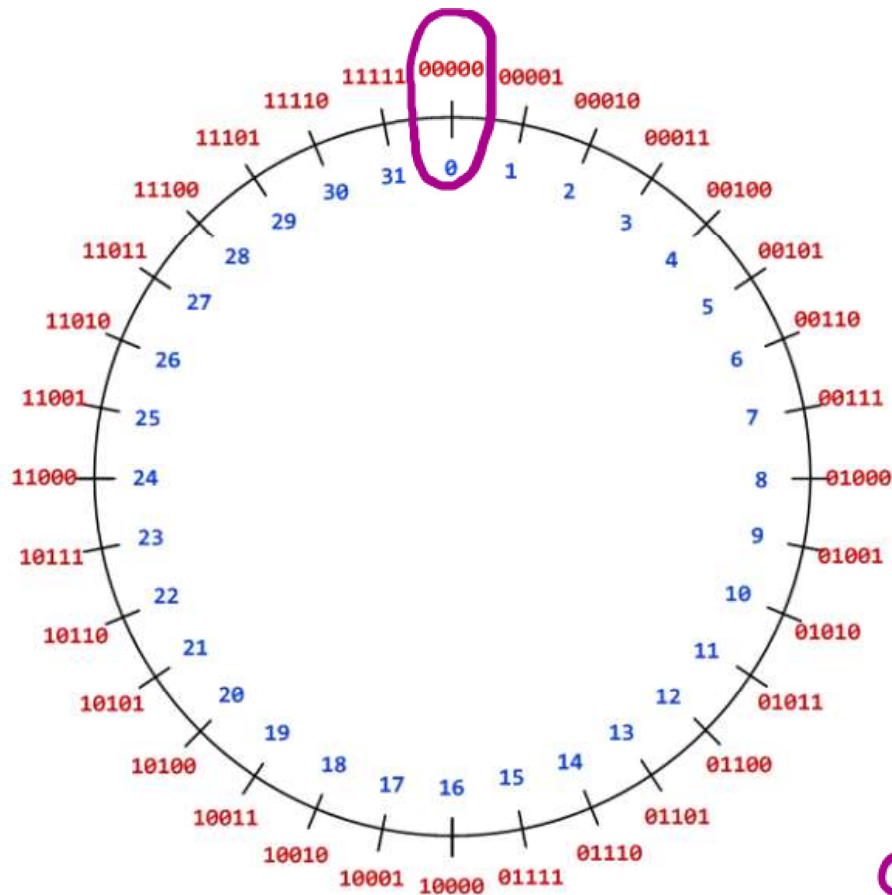
Negative numbers

- Are negative numbers just a number with minus sign in front?





Unsigned Representation



+ don't have any sign

+ Range

5-bit unsigned
binary number

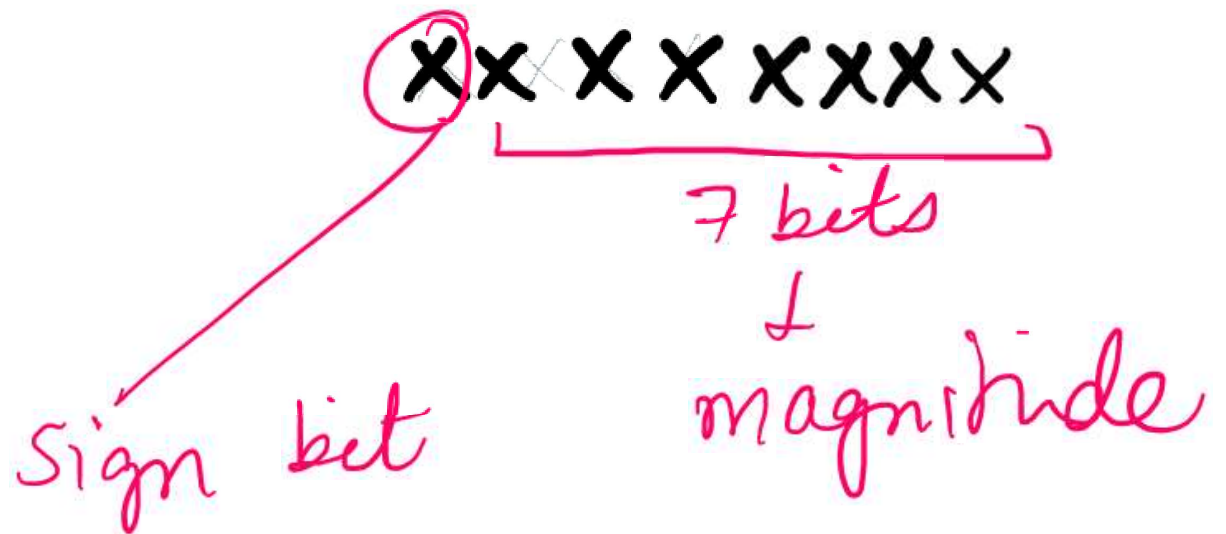
Minimum = 0

Max = 31 = $(2^5 - 1)$

So, 0 to $2^n - 1$ - for n-bit

Signed Representation

① Out of n -bits, Reserve 1 bit as a sign bit



0 → positive
1 → negative

Ex:-

$$\begin{array}{rcl}
 +6 & = & \overset{\text{MSB}}{\textcircled{0}} 0000 110 \quad \text{--- LSB} \\
 -6 & = & \textcircled{1} 0000 110
 \end{array}$$

$\underbrace{\hspace{1.5cm}}$
 sign bit

For a 8-bit integer, we use 1-bit as a sign bit & 7-bit to represent the magnitude of a number, so

Range = $-(2^7-1)$ to (2^7-1) $\xrightarrow{\text{range of values}}$ $\begin{matrix} 0 \text{ to } 2^7-1 \end{matrix}$ } +ve integers
 $\begin{matrix} 0 \text{ to } 2^7-1 \end{matrix}$ } -ve integers

Sign-and-Magnitude:

$$value = (-1)^{sign} \times Magnitude$$

- The most significant bit is the sign.
- The rest bits are magnitude.

▶ Example: in a 5-bit system

▶ $+7_{10} = 00111_2$

▶ $-7_{10} = 10111_2$

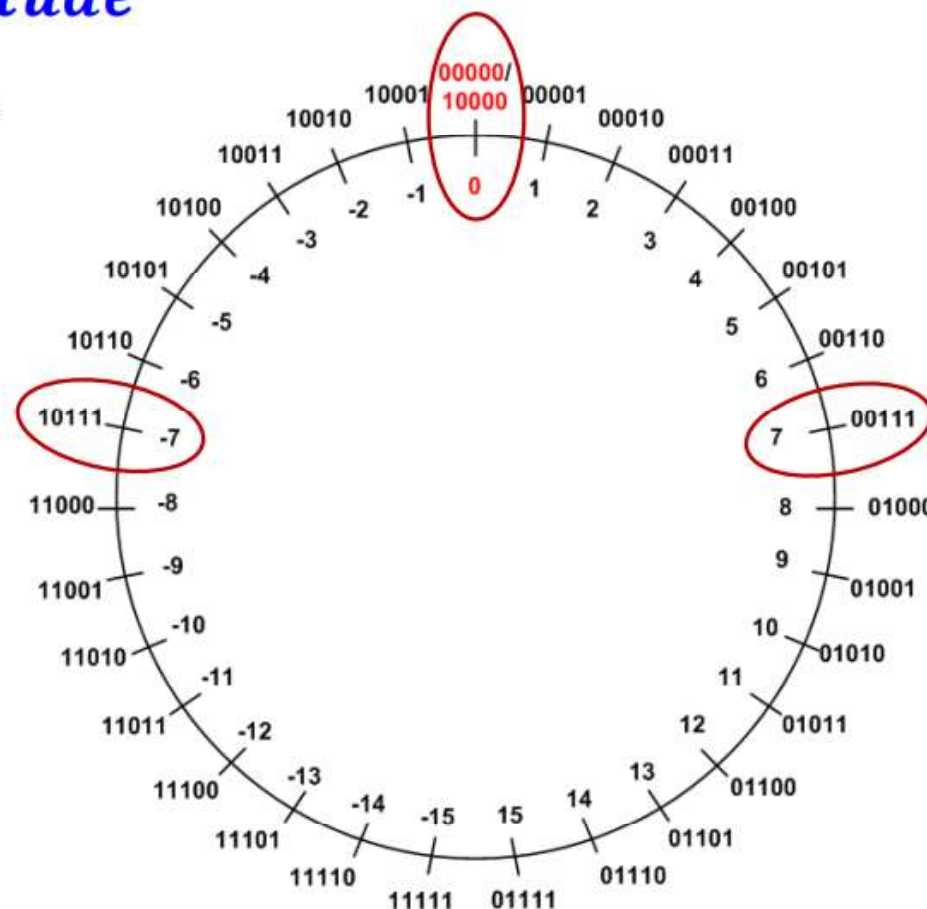
▶ Two ways to represent zero

▶ $+0_{10} = 00000_2$

▶ $-0_{10} = 10000_2$

▶ Not used in modern systems

- ▶ Hardware complexity
- ▶ Two zeros



Signed representation

+0 → 00000000
~~0~~ -0 → 10000000

① Cons:- Zero has 2 representations

+ One representation is sufficient to store zero
+ (Save space & remove ambiguity) if we have one representation for zero.

② Subtraction is not straightforward in signed magnitude representation

Subtraction in Signed Magnitude

$$-5 + (-5) = -10?$$

$$-5 + (\cancel{5}) = 0?$$

$$5 + (-5) = 0?$$

$$\begin{array}{rcl}
 -5 \rightarrow & 1 & 000010 \\
 + \textcircled{5} \rightarrow & 0 & 000010 \\
 \hline
 0 \neq & 1 & 000010 = -10 \text{ (not expected)}
 \end{array}$$

+5 →
 -5 →
 5 →

so we need unique representation for 0.

r 's complement

r = base of the number (radix)

N = whose complement we want to find

n = no. of digits in N .

$$r\text{'s comp of } (N)_r = r^n - N.$$

R's complement

$$r = 10$$

$$N = 7$$

$$n = 1$$

$$10's \text{ complement of } (7)_{10} = 10^1 - 7 = 3 \checkmark$$

R's complement

$$(247)_{10}$$

$$r = 10$$

$$N = 247$$

$$n = 3$$

$$\checkmark \quad 10^3 - 247$$

$$\begin{array}{r} 1000 \\ - 247 \\ \hline \end{array}$$

$$10's \text{ complement of } 247 \rightarrow 753 \checkmark$$

$$9's \text{ complement} \rightarrow 999 - 247 = \underline{752}^{18}$$

r 's complement

$$r\text{'s complement} = (r-1)\text{'s complement} + 1$$

$$r^n - N = \underbrace{(r^n - 1) - N}_{(r-1)\text{'s complement}} + 1$$

② Find $(532)_{10} - (247)_{10}$?

i) Normal

ii) using 2's complement

iii) 8's complement

$$\begin{array}{r} 532 \\ - 247 \\ \hline 285 \end{array}$$

$r=10$

$$\begin{array}{r} 532 \\ + 753 \\ \hline 1285 \end{array}$$

Ignore plus carry

$$\begin{array}{r} 247 \\ 1000 \\ - 247 \\ \hline 753 \end{array}$$

Take Away:-

Ignore the carryover in the result . The remaining Number is your answer .

Carry over → When $n_1 > n_2$ in $n_1 - n_2$

③ Find $(6)_{10} - (3)_{10}$

④ Find $(3)_{10} - (6)_{10}$

10's complement of $-3 = 10 - 3 = 7$

10's complement of $-6 = 10 - 6 = 4$

$\begin{array}{r} 6 \\ \textcircled{-3} \\ \hline 3 \end{array}$	$\begin{array}{r} 6 \\ +7 \\ \hline 13 \checkmark \end{array}$	$\begin{array}{r} 3 \\ \textcircled{-6} \\ \hline \end{array}$	$\begin{array}{r} = 3 \\ +4 \\ \hline 7 \end{array} \rightsquigarrow \textcircled{-3}$
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Take Away :-

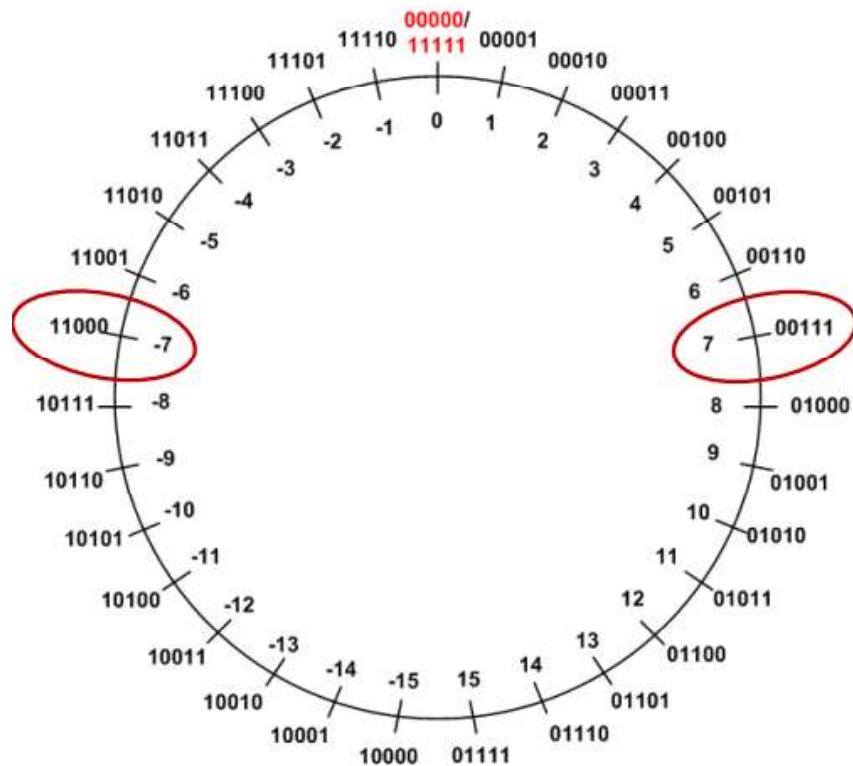
When No carryover $\left[\begin{matrix} n_1 < n_2 \text{ in} \\ n_1 - n_2 \end{matrix} \right]$

{ Take 10's complement of the result and put -ve sign before it. }

Ex:- 10's $\rightarrow$ 10's complement of result
i.e., 7 is $(10-7) = 3 \rightarrow \textcircled{-3} \Rightarrow$

One's Complement ($\tilde{\alpha}$):

$$\alpha + \tilde{\alpha} = 2^n - 1$$



The one's complement representation of a negative binary number is the bitwise NOT of its positive counterpart.

Example: in a 5-bit system

$$+7_{10} = 00111_2$$

$$-7_{10} = 11000_2$$

$$\begin{aligned} +7_{10} + (-7_{10}) &= 00111_2 + 11000_2 \\ &= 11111_2 \\ &= 2^5 - 1 \end{aligned}$$

2's complement

① Problems with signed rep.

- ① 2 rep. of zero's
- ② subⁿ not straightforward

2's complement will solve these two issues

Question using 2s complement?

- 5-5 ?

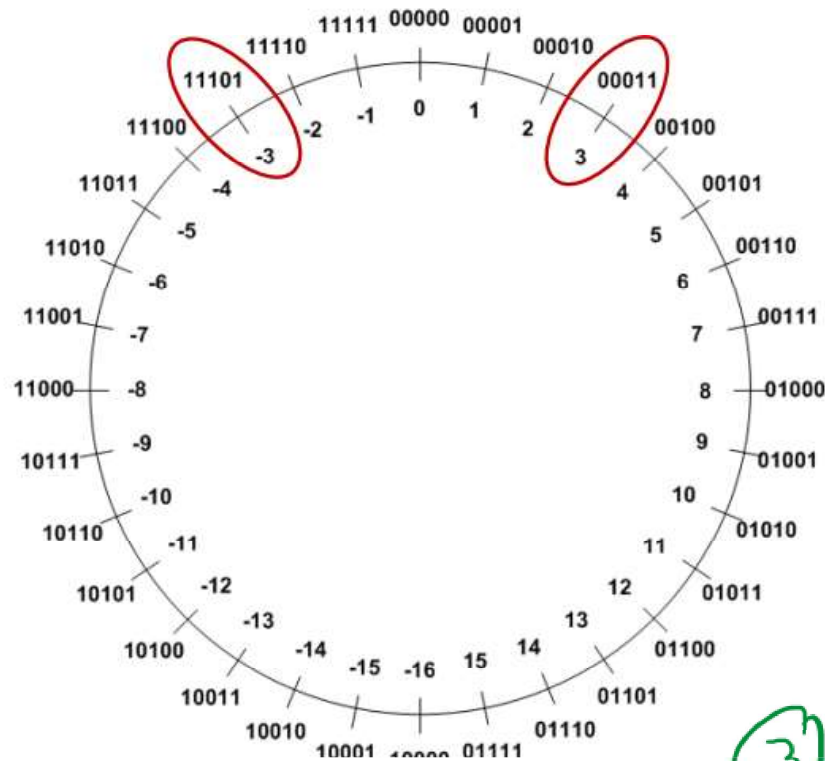
$$\begin{array}{r} 5 \\ -5 \\ \hline 0 \end{array} \rightarrow \text{represent as 8's complement} \\ \text{10's complement}$$

$$-5 = (10' - 5) = \textcircled{5}$$

$$\begin{array}{r} 5 \\ +5 \\ \hline 10 \end{array} \rightarrow \textcircled{\text{Zero}} \checkmark$$

Two's Complement ($\bar{\alpha}$):
 $\alpha + \bar{\alpha} = 2^n$

- ① Removes ambiguity in representation
 ② Provide space to store additional number in place of 0



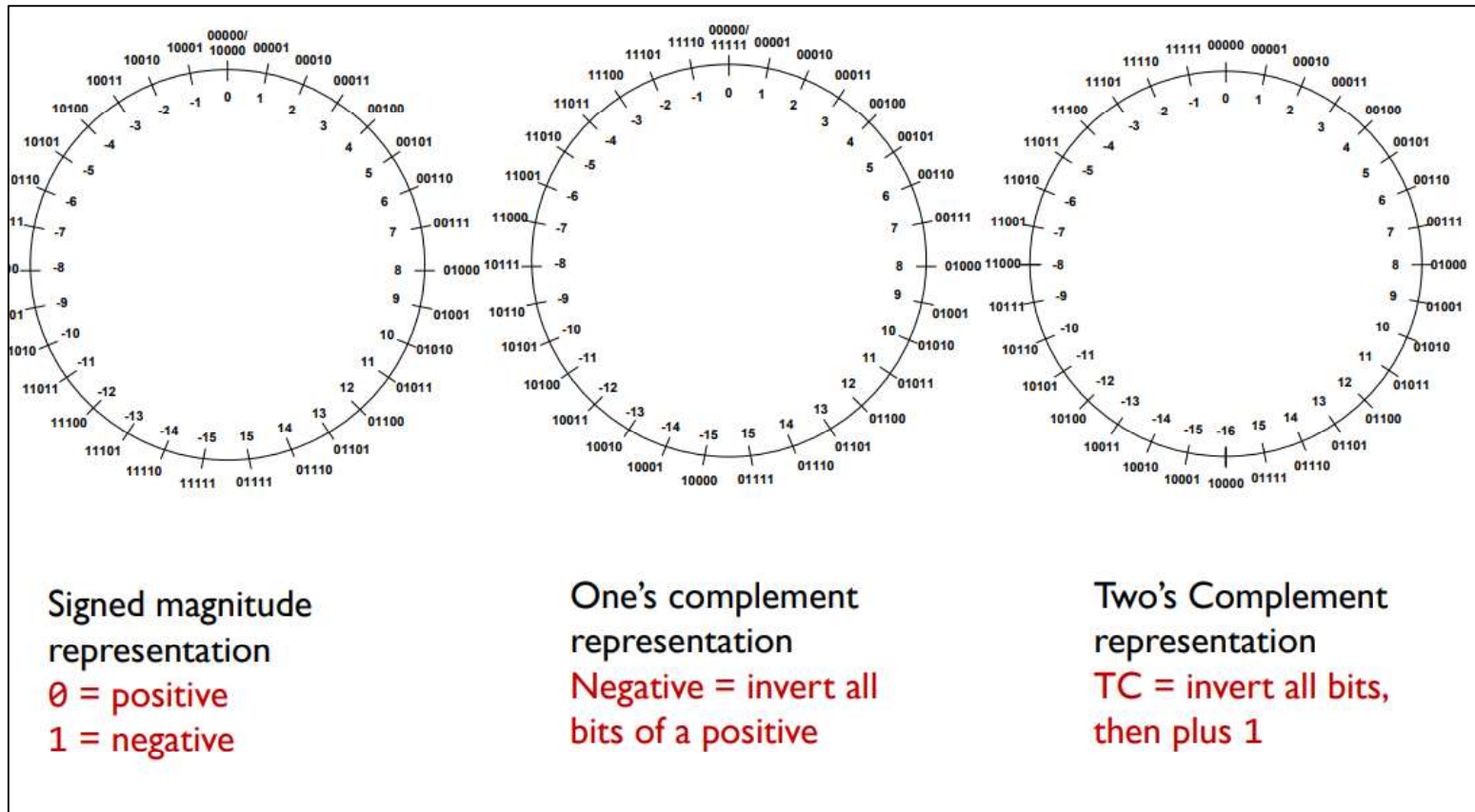
TC of a negative number can be obtained by the bitwise NOT of its positive counterpart plus one.

Example 1: TC(3)

	Binary	Decimal
Original number	0b00011	3
Step 1: Invert every bit	0b11100	
Step 2: Add 1	+ 0b00001	
Two's complement	0b11101	-3

- ③ makes subtraction straightforward.

Comparison



① What is the range of values, Minimum & Maximum values for 8-bit integer in following representations

- 1) Unsigned $\rightarrow 0$ to 255
- 2) ☒ 1's complement $\rightarrow -127$ to 127
- 3) ☒ 2's complement $\rightarrow -128$ to 127
- 4) ☒ Signed $\rightarrow -127$ to 127

Comparison Table

	Sign-and-Magnitude	One's Complement	Two's Complement
Range	$[-2^{n-1} + 1, 2^{n-1} - 1]$	$[-2^{n-1} + 1, 2^{n-1} - 1]$	$[-2^{n-1}, 2^{n-1} - 1]$
Zero	Two zeroes (± 0)	Two zeroes (± 0)	One zero
Unique Numbers	$2^n - 1$	$2^n - 1$	2^n